## Week 4: "Bivariate and Multivariate Analysis"

**Subparts of Exploratory Data Analysis (EDA):-**

- **Bivariate Analysis:-**Bivariate analysis examines the relationship between two variables. It helps in understanding how one variable affects another.

Examples in this project:

- Severity vs Weather Condition

- Severity vs Visibility

- Severity vs Time

- State vs Accident Count

Common Visualizations:

- Bar Chart

- Line Chart

- Scatter Plot

- **Multivariate Analysis:-** Multivariate analysis involves analyzing more than two variables simultaneously. It helps in identifying complex relationships between multiple factors.

Examples in this project:

- Weather + Visibility + Severity

- State + Time + Severity

- Temperature + Weather + Accidents

Common Visualizations:

- Heatmap

- Matrix Chart

- Advanced Dashboard Visuals

**DAX (Data Analysis Expressions):-**

DAX (Data Analysis Expressions) is a formula language used in Power BI to create calculated columns, measures, and calculated tables. It helps in analyzing data and creating meaningful insights. In this project, DAX was used to calculate key metrics and improve data visualization.

Purpose of Using DAX:-

- Create calculated measures

- Generate KPIs

- Perform data aggregation

- Improve dashboard insights

- Support decision-making

Types of DAX Calculations:-

- Measures:- Measures are used to perform calculations on aggregated data.

Examples:

Total Accidents = COUNT(Accident_ID)

Average Temperature = AVERAGE(Temperature)

Total Severity = SUM(Severity)

- Calculated Columns:- Calculated columns create new columns based on existing data.

Example:

Accident Duration = DATEDIFF(Start_Time, End_Time, MINUTE)

- Aggregation Functions:- Common DAX aggregation functions: SUM(),COUNT(), AVERAGE() , MAX() , MIN()

Example:

Total Accidents = COUNT(Accident_ID)

**DAX Used in This Project:-**

- Total Accidents

- Accidents by Severity

- Accidents by Weather Condition & state

- Average Temperature

Benefits of DAX:-

- Helps create KPIs

- Improves data analysis

- Enhances dashboard performance

- Provides meaningful insights

## Slicer Description:-

Slicers were added to the dashboard to provide interactive filtering options for better data exploration. The slicers allow users to filter the visualizations based on different attributes such as state, weather condition, and accident severity. This helps in analyzing specific data segments and understanding patterns more clearly. By using slicers, users can dynamically change the dashboard view and focus on particular conditions, such as accidents occurring in a specific state or under certain weather conditions. This improves the usability of the dashboard and enhances the multivariate analysis.

The slicers used in this dashboard include:

- State Slicer

- Weather Condition Slicer

- Accident Severity Slicer

These slicers help users interact with the dashboard and gain deeper insights into accident patterns and trends.